PakEmis Sector Technical Note TN09

A Reproducible Multi-pollutant Railway Emissions Inventory for Pakistan with Fuel-Balance Closure and Route-Constrained Spatial Allocation

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Abstract

Pakistan's railway sector has existed in public inventories as a single number—and officially, a wrong one. This paper presents a complete, reproducible inventory of the railway module built on the strongest activity datum a sector can have: the operator's own metered fuel consumption, published annually—132 588 thousand litres of diesel in the FY2023–24 reference year, 0.394 Mt CO_2e in all, of which 0.389 Mt is strict locomotive traction at IPCC 1A3c and 0.005 Mt is the measured non-traction claimant, booked at its own category pending an unresolved 1A2/1A4c end-use split. Around that control it adds the first railway air-pollutant layer for Pakistan (EMEP/EEA Tier 1: 5.87 kt NO_x, 0.154 kt $PM_{2.5}$, black carbon as the chapter's declared fraction, and SO_2 as an explicit sulfur-case ladder spanning 10 ppm to 10 000 ppm); a twelve-year unbroken fuel series with a quinquennial history to 1950; a parametric-plus-structural uncertainty system; an independent two-source diesel cross-check against the petroleum industry's own Railways customer class. A route-constrained allocation over an audited OpenStreetMap network is retained as an ODbL-derived methodological alternative. The primary release carries its compatible administrative allocation and retains unsupported district/grid placement as unallocated; these products must not be described as the same observed spatial account.

The refutation that motivated this paper's first version stands, strengthened and matched by year. The First Biennial Update Report's 4.06 Mt for FY2017–18 is 8.7 times the 0.466 Mt the identical method yields on that year's metered fuel; and the error is not historical—Pakistan's current Common Reporting Tables book 56 609 TJ for railways, 12.6 times the metered fuel, with the *road* non-CO_2 factors visible in the filing's own implied-emission-factor cells. Every table in this paper is reconciled row by row against the engine's outputs by tests, and a hashed-source manifest closes each run.

Keywords: railway emissions, diesel traction, air pollutant inventory, national inventory verification, emission factor tables, route-constrained allocation, IPCC 2006 Guidelines, EMEP/EEA Guidebook, Pakistan

PakEmis Sector Technical Note TN09. Technical documentation of one sector module of the PakEmis release identified below, deposited with the data as part of the documentation record. The national data-description article (`paper_national/`) is the only document that states the national totals as its own result.

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Release provenance. This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

Release version	`1.0.0rc69`
Commit	`f3f7faec9d6a`
Reporting period	July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30)
Reference	S_TAN (household survey; TAN nitrogen)
FY2023–24 total	444.61 Mt CO_2e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN
Release manifest	`c5ca5795eb24666c` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

1. Introduction

The railway module is the smallest fuel-burning sector in Pakistan's national greenhouse-gas inventory and the best measured. Pakistan Railways is one state operator. It meters its own fuel consumption and publishes the annual total in a Year Book

alongside the traffic that consumed it. There is no fleet model, no split between competing operators, and no need to allocate a national fuel supply between uses. The activity datum is one number a year, and everything else in a railway inventory—gases, air pollutants, geography, history—is that number decomposed.

That is the design principle of this paper: the metered total sets every magnitude, and subsidiary information sets only distributions, each reconciling back to the control exactly. It is the architecture this inventory's agriculture sector applies to six IPCC categories with dispersed activity; rail is the sector where it can be applied almost without assumptions, because the control is a fuel meter rather than a statistical model.

The paper makes five contributions. First, a corrected national railway greenhouse-gas series, FY2014–FY2025 unbroken, with a quinquennial fuel history to 1950 showing the sector once emitted 5.8 times its present total on furnace-oil traction. Second, the first multi-pollutant railway inventory for Pakistan, on the EMEP/EEA Guidebook's railway chapter with sulfur handled as declared Pakistani cases rather than European defaults. Third, an independent cross-source check on the operator's own fuel account: the Year Book's metered purchases against the petroleum industry's Railways customer-class total, two organisations counting the same physical fuel by different routes. It replaces an internal deduction from road transport's control, which the re-anchoring of that sector onto customer classes made structurally unnecessary. Fourth, a route-constrained spatial inventory on the audited OpenStreetMap rail network, retained as a separately licensed alternative. The primary release preserves its compatible administrative support and unallocated remainder. Fifth, a matched-year evaluation against every official statement of this category—and that evaluation is the paper's sharpest result.

Pakistan's First Biennial Update Report put railways at 4.06 Mt CO_2e for FY2017–18. This inventory, run on the operator's metered fuel for that same fiscal year, gets 0.466 Mt—a factor of 8.7—and 0.394 Mt for the FY2023–24 reference year (module total; 0.389 Mt of it strict 1A3c traction under the re-based reference). The gap is not a year effect: the two years' fuel differs by 18.0 %. And it is not history: the current Common Reporting Tables carry 56 609 TJ of railway activity, 12.6 times the metered fuel, with implied CH_4 and N_2O factors of 3.9 kg/TJ each—the *road* table's values, readable in the filing's own cells—where the rail table says 4.15 and 28.6. Section 6 works the disagreement through against the carbon in the fuel, the traffic that burned it, an independent global inventory and the Auditor-General's litre-level findings; the factor-table trap itself is the subject of Section 5.

The paper records these disagreements rather than resolving them silently, on the view that an inventory which disagrees with its country's submission without saying so is worse than one that disagrees openly—and every number printed here is held to that standard mechanically: tests reconcile each table row by row against the engine's outputs, and a manifest hashes every primary source each run reads.

2. Related Work

2.1. Rail emissions estimation

Railway greenhouse-gas accounting divides on whether traction fuel is metered. Where an operator publishes its own fuel consumption, as in most national rail systems, IPCC's Tier 1 is already close to a measurement: the uncertainty in the estimate is essentially the uncertainty in the emission factor. Where fuel is not published—typically for fragmented or private freight networks—inventories model traction energy from gross tonne-kilometres and specific energy consumption, and the resulting figures carry the uncertainty of a fleet model.

Pakistan sits firmly in the first case, which is why this paper spends no space on traction modelling and considerable space on what to do when a published national figure disagrees with the metered one.

2.2. Inventory verification against national submissions

Independent verification of national submissions is normally attempted at the level of totals, using atmospheric inversions or independent global products such as EDGAR (Crippa et al., 2025), and it usually produces a disagreement that cannot be adjudicated: two models differ, and neither is a measurement. This inventory's own fugitives sector is an example, where a factor of 1.3 against EDGAR settles nothing because both estimates choose within the same published range.

Railways is different in kind. The disagreement here is against an activity figure that has a physical bound—the carbon in a quantity of diesel a state operator reports consuming—and a second, independent bound in the traffic statistics published in the same document. A submission can be refuted on arithmetic when the arithmetic is this short, and that possibility is rarer in inventory work than the volume of comparison studies suggests.

2.3. Emission-factor tables as a failure mode

The literature on inventory quality concentrates on activity data and on factor values. It says much less about factor *selection*—which of several published tables applies to a given category—even though the guidelines themselves contain several near-duplicate tables that differ only in their non-CO_2 rows.

This inventory has now documented three instances: residential fuel against the power table, power against several, and rail against road here. The common structure is that the CO_2 row is identical across tables because CO_2 depends only on the carbon in the fuel, so the dominant number in the category is unaffected and every obvious check passes. Section 5 treats the rail case and Section 9 the general one.

3. Study Area and Data

3.1. One operator, one number

Pakistan Railways operates the country's mainline rail network as a state enterprise. Its Year Book reports fuel consumption by product, traffic by division, and train-kilometres

by service class. The datum is what the source calls it: the table is headed *Fuel Consumption*, and the narrative states that the quantities "comprise the entire fuel consumed on the Railways, i.e. on locomotives, power houses, water pumps, workshops etc." This inventory adopts the Tier 1 assumption that fuel consumed in a fiscal year equals fuel burnt in it; no depot-stock or procurement-timing correction is published, and none is attempted. The reference year here is FY2023–24: the 2024–25 edition is a page-image scan whose own Legal Opinion page declares its data provisional pending audit, so its year enters the series flagged provisional rather than as the reference.

Stream	TJ	% of energy
Diesel	4817.6	99.933
Furnace oil	3.2	0.067

Table 1: Railway module fuel, FY2023–24. Diesel is 132 588 thousand litres—the metered total, of which 1599 thousand litres is the measured non-traction share; furnace oil is a rounding error carried for the reason given below.

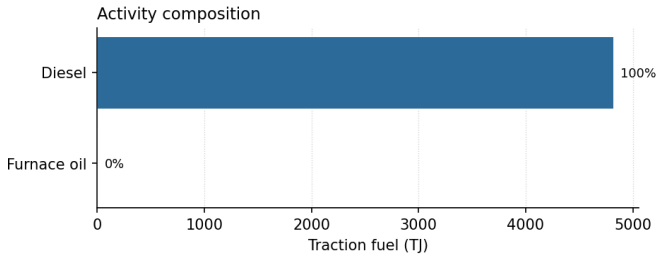


Figure 1: The energy basis. One stream is the sector.

3.2. Electric traction: none since about 2010, and none in scope

Recent Year Book editions report Electric Energy as NIL, and the historical table behind that figure says when it became true. Electric traction energy is non-zero from the 1985–90 quinquennium (37.4 GW h per year on average) and declines through 2005–2010 (6.1 GW h) before going NIL from the 2010–2015 row. The Lahore–Khanewal electrification—the operator's only electrified section—therefore ran into the late 2000s, not the 1970s as this paper previously stated; the mainline network has been diesel-only since roughly 2010, which covers every inventory year here. The sector consequently has no scope-2 stream, unlike residential fuel where the electricity boundary has to be argued.

The scope is the Pakistan Railways mainline system's direct combustion. Urban electric rail outside the operator—the Lahore Orange Line metro—has zero direct 1A3c combustion emissions by definition; its electricity-related emissions belong to power generation, and the boundary is recorded in the completeness assessment rather than left implicit in a NIL row.

3.3. Two things the source does badly, both recorded

Five editions are scans, and it turned out not to matter. The 2017–18, 2018–19, 2019–20, 2021–22 and 2024–25 editions are page images with no extractable text layer, and this paper previously reported FY2018 and FY2019 as absent from the time series for that reason.

That was true about those editions and wrong about the data. Every edition's fuel table carries a historical run, not just the current year, and the 2020–21 edition has a text layer: it reports FY2015–16 through FY2020–21 annually. Both missing years were one edition later the whole time. They are now in the series—156 396 and 159 927 thousand litres—recovered rather than interpolated, and the series runs FY2014–FY2025 unbroken.

What makes them trustworthy is that the same table reproduces FY2016, FY2017 and FY2021 exactly against figures this inventory already held from other editions. A test asserts that correspondence, and a second asserts the series is contiguous, because the honest reason to leave a hole in it has gone.

Furnace oil changed unit labels without announcement, and the source itself settles the basis. Editions through 2020–21 label the row "Tonnes" and editions from 2022–23 label it "Liters/thousand," with no note explaining the change. The 2023–24 edition resolves it against itself: its detailed fuel table and the narrative on the same page state the quantity in metric tonnes, and the summary table's unit column is demonstrably unreliable (Section 10). The basis is tonnes, and that is what the activity file carries. The stream is under 0.001 Mt CO_2e on either reading, so nothing material turns on it, and a test pins that indifference by recomputing the total under both interpretations. The unit travels with each row in the activity file rather than being inferred from the year, so any future correction is a data edit and not a code change.

Furnace oil is carried at all for two reasons: IPCC Chapter 3 §3.4.1.4 warns specifically against omitting non-diesel traction fuels, and an unexplained unit change in a government series is worth recording where the next person will find it.

3.4. The traffic statistics

The same Year Book reports 26 885 thousand passenger, mixed and other coaching train-kilometres and 6237 thousand freight, 33 122 thousand in total. These do not enter the estimate. They are carried as a parameter because they permit the fuel figure to be checked against the traffic that consumed it, which is the evidence behind Section 6.

4. Methods

4.1. The calculation

Litres to tonnes to energy to emissions, once per fuel and per gas:

$$E_{f,g} = Q_f \cdot \rho_f \cdot \mathrm{NCV}_f \cdot \mathrm{EF}_g, \tag{1}$$

where Q_f is the published quantity, ρ_f the density needed to reach a mass basis, NCV_f the net calorific value, and EF_g the IPCC rail factor for gas g. There is nothing else in the sector.

The density matters more than it looks. Pakistan Railways reports litres and every IPCC factor is per unit of energy, so a density is unavoidable, and it is taken from Pakistan State Oil's own high-speed-diesel material safety data sheet—specific gravity 0.83 to 0.86 at 16 °C, midpoint 0.845 kg L^{-1}—rather

than from a generic international default. That spread is the entire uncertainty in the step, and at 2 % it is the smallest term in the sector.

4.2. The furnace-oil rows

Furnace oil takes its CO_2 factor from the fuel's own carbon content in Volume 2 Chapter 2 Table 2.2, because Table 3.4.1 tabulates only diesel and sub-bituminous coal. Its CH_4 and N_2O factors are the diesel rail values, following §3.4.1.4, which directs that other traction fuels burnt in the same engines be treated with the same emission properties. Both choices are recorded on the factor rows rather than inferred from the code.

4.3. Fossil methane

Diesel and furnace oil are fossil fuels, so their CH_4 takes AR5's fossil value of 30 rather than the biogenic 28—the same treatment as road transport, power and aviation, and the opposite of waste. A test asserts the CO_2e of every CH_4 row against 30 exactly, because the two values differ by seven percent and nothing in the output distinguishes them.

4.4. The air-pollutant layer

Air pollutants take the EMEP/EEA Guidebook's railway chapter (Norris and Ntziachristos, 2023) at Tier 1: factors per tonne of fuel, on the identical fuel-mass chain as the greenhouse gases, in kilotonnes of native mass that never enter a CO_2e total. Black carbon is the chapter's declared fraction (0.65) of $PM_{2.5}$—a component of the particulate mass, never an addition to it—and the chapter's one defect is recorded rather than repaired: its printed 95% intervals for the three particulate rows do not bracket their own central values, so those rows carry the printed bounds verbatim and a conservative ±100 %.

SO_2 is stoichiometry, not a factor: twice the fuel's sulfur mass. The chapter says operator sulfur values are preferable to defaults; Pakistan Railways publishes none, and Pakistani diesel spans three orders of magnitude—10 ppm Euro-V imports, mandatory since January 2021, against the 500 ppm pre-Euro-V specification and the older specification's 1 % ceiling that covers most of the historical series. So SO_2 ships as an explicit case ladder with the 500 ppm specification as the declared reference, not as one number pretending to precision the fuel supply does not have.

A Tier 2 diagnostic accompanies the reported Tier 1 layer: fuel split among line-haul passenger, line-haul freight and shunting (railcars are not occurring—workings are Nil in every recent year), rescaled to the metered control exactly, as the chapter itself instructs. Published train-kilometres divide the line-haul fuel; the shunting share is a declared five-percent assumption, derived from the chapter's own duty rates and gap-registered, because no shunting activity is published in any form. What the diagnostic shows is the duty-cycle sensitivity: a nearly-all-line-haul fleet implies Tier 1 understates NO_x by roughly a fifth and overstates fine particulate mass by about a sixth, and neither layer is silently swapped for the other.

4.5. The diesel closure against road transport

Pakistan's national high-speed-diesel supply figure covers road, rail, agriculture and captive generation together. While the road transport sector anchored on that single national figure, every litre Pakistan Railways burns was at risk of being counted twice—once here and once as road diesel—and the road sector deducted this module's diesel, litre for litre, by calling it rather than by holding a copy of the number.

That deduction has been retired, and this section records why rather than describing a mechanism that no longer runs. Road transport is now anchored on the petroleum industry's own customer classes, in which Railways is a separate row from Road. The double-count is excluded by construction at the source, so subtracting the railway account from the road control would now remove the same fuel twice.

What the railway fuel account does instead is better. Freed from being an internal correction, it is an *independent cross-source check*: the operator's own metered purchases, from the Year Book, against the petroleum industry's Railways customer-class total, two organisations counting the same physical fuel by different routes. The quantity itself is unchanged— the module's metered diesel is 112.04 kt, traction and non-traction together—and only what the inventory does with it has moved. An internal deduction can only be arithmetically consistent; an independent comparison can be wrong, which is what makes it worth reporting. Section 6 gives the comparison.

The general point outlives this instance. A closure that becomes structurally impossible has been superseded by better source data rather than broken, and the right response is to delete the correction and say so—not to keep a compensating adjustment running against a control that no longer needs it.

One deduction from road transport's controls does survive, and it is not this sector's: household generator petrol, owned by the residential module. A test asserts that the railway account contributes nothing to it—Pakistan Railways burns no petrol, and deducting from a fuel the operator does not use would be a second error correcting the first.

4.6. Spatial support and the ODbL route alternative

The following route reconstruction is an ODbL-derived alternative, not the primary package's licensed grid. The primary product retains its compatible administrative support and records unsupported finer placement as unallocated. A source-specific physical rail release must not inherit an unrelated grid merely to eliminate that remainder.

Railway emissions can occur only along a railway, so the allocation is constrained to one. Line-haul fuel runs along the OpenStreetMap-derived network—7574 active line-kilometres, 97 % of the Year Book's 7791 route-kilometres—divided into corridors; shunting fuel sits at the seven divisional yards, weighted by their published wagon loadings; and each of the three service streams takes its fuel share from the Tier 2 split, on a surface that reconciles to the national control exactly at national, provincial and district level on every run.

The honest split of that construction: the network geometry, the yard locations and the route-kilometre audit

are data; the corridor service weights are *declared judgment*, stated basis-by-basis on their data rows, because the operator's public timetable interface refuses unauthenticated access and no service-frequency statistics are published. The previous proxy—freight wagons loaded by division, which credited Sindh with fuel burnt hauling Karachi-origin freight up-country and allocated passenger operations (four times the freight train-kilometres) by freight geography—is retained as a structural alternative, and both splits are published. The national total is identical under either; the engine reads the transcribed share row, and a test pins that row against a fresh spatial computation so the two cannot drift apart.

5. Emission Factors

Gas	Rail (3.4.1)	Road (3.2.1/3.2.2)	Ratio
CO_2	74 100	74 100	1.00
CH_4	4.15	3.9	1.06
N_2O	28.6	3.9	7.33

Table 2: IPCC 2006 defaults for diesel, kg per TJ, from Volume 2 Chapter 3 (IPCC, 2006). The CO_2 rows are identical because CO_2 depends only on the carbon in the fuel; the N_2O rows are not.

5.1. Why rail N_2O is 7.3 times road

Large slow-speed locomotive engines run lean at high load, which is the N_2O-forming regime. IPCC derives the rail values in Table 3.4.1 from EEA measurements at 0.35 L per brake-horsepower-hour for a 4000 HP locomotive, and the 28.6 kg TJ^{-1} value carries a 95% interval of 14.3 kg TJ^{-1} to 85.8 kg TJ^{-1}. The road value of 3.9 kg TJ^{-1} describes a different engine population running a different duty cycle.

5.2. What the wrong table would do

Substituting the road N_2O factor would understate railway N_2O by 86 % and the sector total by about 8 %, from 0.394 to 0.363 Mt. That figure is not left as a claim in a docstring: the runner computes it and publishes it as a diagnostic row in the sector's totals file, so a reader can see the size of the choice rather than take it on trust.

5.3. The trap, caught in the current filing

This is no longer only a hypothesis about how the BUR went wrong. Pakistan's current Common Reporting Tables carry, in the implied-emission-factor cells of the railway category's own liquid-fuels row, CO_2 at 74.10 t/TJ, CH_4 at 3.90 and N_2O at 3.90 kg/TJ—the road table's non-CO_2 pair, on rail, in the official submission of record. The cell-audited extraction behind that statement reproduces the filing's arithmetic from its own workbook before comparing anything to it, and a test pins the finding so that a corrected future submission fails the pin and forces this paragraph to be rewritten rather than left standing.

5.4. Why this is a trap and not a mistake

The failure is silent in every direction a compiler would normally look. Both tables are in the same chapter of the same volume. Both are labelled for transport. The CO_2 row—which is 90 % of the category—is byte-identical, so a total computed with the wrong table still reconciles against fuel, still matches a carbon-balance check, and still passes any comparison run on CO_2 alone. Only the N_2O row moves, and only by an amount that looks like ordinary uncertainty in a category this small.

Four tests hold the choice in place. One asserts that the cited source contains "Rail." One asserts that rail N_2O exceeds five times the road value, so a silent substitution of the transport rows fails immediately. One asserts that rail CH_4 exceeds road CH_4. And one asserts that N_2O is more than 5 % of the sector total—because a pinned factor that nothing depends on is decoration, and the guard is only worth having while the choice still moves the answer.

6. Validation

6.1. A national submission, refuted twice

Pakistan's First Biennial Update Report reports railways at 4.06 Mt CO_2e for FY2017–18. That fiscal year is inside this inventory's series, so the primary comparison is matched-year: the identical method, GWP basis and scope, applied to the 156 396 thousand litres the operator reports for FY2017–18, gives 0.466 Mt—a factor of 8.7 below the submission in its own inventory year. The reference-year result is reported separately: 0.394 Mt module total for FY2023–24 (0.389 Mt strict 1A3c). A year effect is ruled out by the fuel itself: FY2017–18 diesel is 18.0 % above FY2023–24, not a factor of ten in either direction.

The carbon route. 4.06 Mt of CO_2 requires the combustion of roughly 1280 kt of diesel. Pakistan Railways reports consuming 132 kt in FY2017–18, and 112 kt in the reference year. The submitted figure needs nearly ten times the fuel the operator's own accounts record for the same year, and no accounting treatment of a fuel account produces carbon that was not in the fuel. A test bounds the sector total between 0.2 and 1.0 Mt for this reason, so a future revision that drifted toward the submitted value would fail rather than pass quietly.

The traffic route. The second check is independent of the fuel account, because it uses a different table in the same publication. Pakistan Railways ran 33.1 million train-kilometres in FY2023–24. The metered consumption works out at 4.0 L per train-kilometre, which is where a diesel locomotive hauling mixed passenger and freight traffic belongs. The fuel implied by the submitted figure, spread over the same traffic, would be 46 L per train-kilometre—an order of magnitude above what any locomotive consumes. A test pins the intensity between 2 and 10 litres per train-kilometre.

Two routes, two published tables, one conclusion. The metered consumption, the traffic statistics and the carbon content of diesel all support the lower figure, so it is the one reported here.

6.2. *Why this is recorded rather than reconciled*

The alternative to publishing this disagreement is to adopt the submitted value, or to split the difference, or to say nothing. All three would be worse.

Adopting it would require inventing ten times the fuel Pakistan Railways reports consuming in the submission's own year. Splitting the difference would produce a number supported by no source at all. Saying nothing would leave a reader who compares this inventory with Pakistan's UNFCCC submission to discover a factor of ten unexplained, and to have no way of telling which figure to trust.

What is defensible is to state the disagreement, show the arithmetic on both routes, put the checks in the test suite so they run on every build, and let the reader see the evidence. This is the same standard applied in this inventory's fugitives sector to a disagreement with EDGAR that could *not* be resolved—and the contrast is the point. There, two models chose differently within one published range and the comparison settled nothing. Here, one figure is outside what the fuel can produce.

6.3. *Every official statement of this category, matched by year*

Source	Year	Official	This work	Ratio
INC (derived)	FY1993–94	0.557	0.633	0.88
BUR-1	FY2017–18	4.060	0.466	8.7
TNC	CY2021	*railways not separately reported*		
BTR-1 (narrative)	CY2021	*railways not separately reported*		
CRT (= BTR-1 annex)	CY2022	4.259	0.368	11.6

Table 3: Every official statement of this category, Mt CO_2e, each compared in its own inventory year by the identical method. The INC row is derived from its stated share (3% of 18 584.4 Gg) and compared against the quinquennial history layer's 1990–95 indicative value—a magnitude check, not a point match. The TNC and the BTR-1 narrative report 1.A.3 as one aggregate line; the BTR-1's railway statement exists only in its CTF annex, which is the CRT row. The CRT total is stated on the filing's own GWP basis, its calendar year straddling two fiscal years (own value: their mean).

The table's arc is the paper's history of the category in five rows. The INC-era inventory was *sane*: its derived railway figure sits within 11.9 % of this inventory's independent reconstruction from the Year Book's own fuel history, furnace-oil traction included. The order-of-magnitude error entered the record with BUR-1, persists in the current CRT, and the two documents in between decline to state a railway figure at all—which is the procedural gap made visible: a submission that published subsector activity alongside its estimate would be auditable, and the one time it was (the CRT), the audit takes one worksheet.

The CRT comparison adds what the BUR comparison could not: mechanism. The filing's implied-factor cells show the road table on rail (Section 5), and its activity—56 609 TJ, about 1316 kt of diesel—implies eleven to thirteen times the fuel the operator metered in the overlapping years. The order-of-magnitude error and the factor-table error are distinct, visible separately in the filing's own arithmetic, and both are pinned by tests against the extracted cells.

6.4. *An independent global inventory, and what it misses*

ELK, the one open global inventory with separate passenger- and freight-rail layers, carries a non-zero Pakistan rail signal: 172 kt of CO_2 for 2019, country-masked, against this inventory's 404 kt for the same calendar year—0.43 times, a factor-of-two disagreement rather than the officials' factor of ten, and evidence that the global model did not ingest the BUR. Its composition, however, is inverted: 165 kt on freight across a thousand grid cells against 7 kt of passenger rail on five cells, for an operator whose traffic is four-to-one *passenger* by train-kilometre. A fleet model without national activity data reconstructs the wrong railway; the operator's two published numbers—fuel and train-km—are what it lacks. The comparison is a benchmark, never a calibration target, and the extraction states its hypotheses before computing.

6.5. *The auditor's litre-level corroboration*

The one-operator limitation is real: the Year Book supplies both the fuel account and the traffic that checks it. The Auditor-General's railway audits are the independent institution, and their findings corroborate the meters precisely by documenting control failures: 710 143 litres consumed against a 316 664-litre quota across six formations in FY2023–24; 393 554 litres above per-class trip rations in an earlier year; 23 250 litres pilfered from stabled locomotives and detected against the tank record. Quota breaches resolved at the litre level per formation are only possible where fuel issue is metered. The audited excesses are fractions of a percent of the national figure: fuel *control* is weak, fuel *measurement* is not, and it is the measurement this inventory rests on.

6.6. *Intensity, over time and against the world*

The EMEP chapter's verification guidance is fuel against distance and traffic. Pakistani rail runs at 4.0 to 5.1 litres per train-kilometre across FY2022–FY2025, with FY2023–24 at 4.0; freight moves at roughly 8 g CO_2 per tonne-kilometre and passengers at under 10 g per passenger-kilometre—the heavy-load end of world rail, consistent with long, full trains. The World Bank's UIC-sourced series, institutionally independent of the Year Book, corroborates the passenger-kilometre magnitude. Every quotient is diagnostic; none feeds the inventory.

6.7. *Internal checks*

Three further properties are asserted on every run. Diesel exceeds 99 % of traction energy, so a corruption of the furnace-oil row cannot move the sector. The sector total is indifferent to the furnace-oil unit ambiguity, recomputed both ways and agreeing to better than 0.001 Mt. And the provincial allocation reconciles exactly against the national figure.

The road-transport comparison of Section 4.5 is enforced by test on every run: the two rail figures—the Year Book's metered 112.04 kt and the OCAC Railways customer-class row at 107.93 kt, two unrelated reporting chains 3.8 % apart—must agree within 5 % and must *not* be equal, because equality would mean one had been copied from the other; the matched-year pair must beat the two-year bracket; the road diesel control must

equal the OCAC Road row exactly, with no rail subtraction anywhere; petrol carries no rail contribution; and rail's share of national high-speed diesel stays between 0.2 and 5 %—because if that ratio ever left the range, one of the two national figures for the same fuel would have broken.

7. Results

7.1. The sector

Pakistan's railways emitted 0.394 37 Mt CO_2e in FY2023–24 on AR5 metrics, or 0.395 47 Mt on AR6—and the reference itself books the measured split the Year Book's detail table provides: 0.389 39 Mt of locomotive traction at 1.A.3.c, and 0.004 98 Mt of workshop, power-house, pump and track-machinery fuel at its own category (1A2/1A4c, the split below the one published aggregate unresolved), rather than declaring the latter IE inside the traction code.

Stream	Gas	Gg	Mt CO_2e	±%
Diesel (traction)	CO_2	352.678	0.35268	5.4
	N_2O	0.136	0.03607	150.1
	CH_4	0.020	0.00059	130.1
Furnace oil (traction)	CO_2	0.047	0.00005	30.1
Non-traction, all fuels	all		0.00498	
1A3c locomotive traction			0.38939	
Non-traction (1A2/1A4c)			0.00498	
Module total			**0.39437**	

Table 4: Railway module streams, FY2023–24. Two furnace-oil rows below 0.0001 Mt are omitted from the table and included in the total. The non-traction row's allocation between 1A2 and 1A4c is unresolved; it books at its own category pending a per-use split.

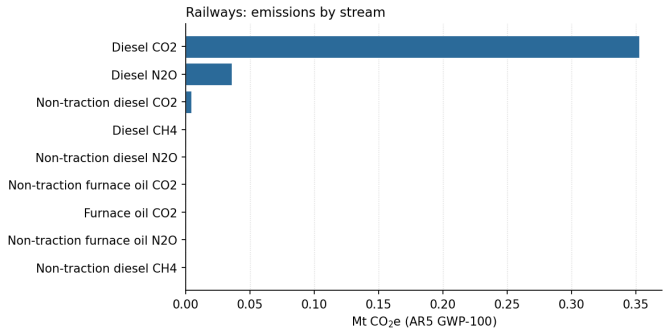


Figure 2: Emissions by stream. CO_2 is 90.6 % of the category and N_2O is 9.3 %, which is the entire reason the factor-table choice matters.

The uncertainty structure is unusual for this inventory. Diesel CO_2 at ±5.4 % is the best-constrained stream in the whole national estimate—a single state operator's metered consumption against a factor whose 95% interval spans 3 %—while N_2O at ±150.1 % is among the worst. The sector is simultaneously the most accurately measured and, in its second gas, the least confidently factored.

Case	Mt CO_2e
Rail table (3.4.1), reported	0.39437
Road table (3.2.2) substituted	0.36282

Table 5: What the sector would report on the road N_2O factor. The gap is 8.0 % of the category and 100 % of the reason the table choice is pinned by a test.

Pollutant	kt, FY2023–24	±%
NO_x	5.8749	65.2
CO	1.1996	61.2
NMVOC	0.5213	65.2
TSP	0.1704	100.1
PM_{10}	0.1614	100.1
$PM_{2.5}$	0.1536	100.1
BC	0.0998	104.5
SO_2	0.1176	100.1
NH_3	0.0008	57.2

Table 6: Railway air pollutants, EMEP/EEA Tier 1 on the metered fuel; native mass, never CO_2e. BC is a component of $PM_{2.5}$; SO_2 is the 500 ppm reference case of a 0.008 kt to 2.25 kt sulfur ladder.

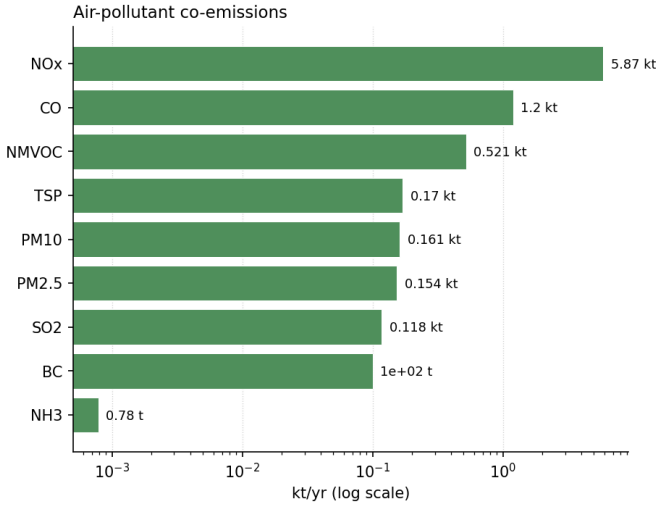


Figure 3: The air layer on a log axis: NO_x in the kilotonnes, the particulate family in the low hundreds of tonnes, black carbon just under 100 tonnes.

7.2. The factor-table diagnostic

7.3. The air-pollutant layer

For scale: railway NO_x at 5.9 kt is a small national term, but it is emitted in a hundred-metre-wide corridor through the centres of Pakistan's largest cities, which is what the route-constrained surfaces of Section 4.6 exist to place. SO_2 is the one pollutant whose national value is genuinely open: on Euro-V fuel throughout it would be 0.008 kt, on the historical 1 % specification 2.25 kt—a 287-fold span that only an operator fuel-quality certificate can close.

7.4. The series, and what it shows about Pakistani rail

Three features, one of them only visible now the gaps are filled. The trend is upward—two fifths more fuel across the twelve-year series, against a national inventory in which most transport categories also grew. The series is volatile, dropping

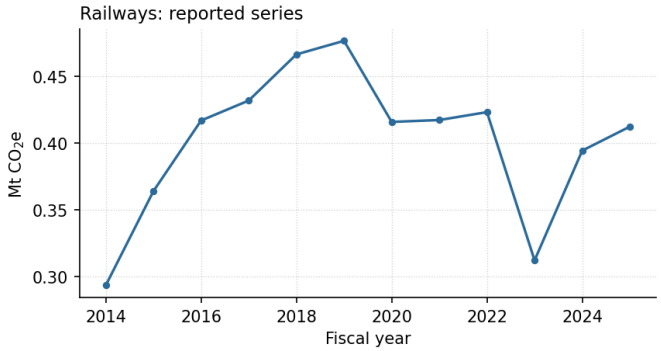


Figure 4: FY2014–FY2025, unbroken. Emissions rise from 0.294 to 0.412 Mt as diesel consumption rises from 98 212 to 138 673 thousand litres, peaking at 0.477 Mt in FY2019; the reference year FY2023–24 stands at 0.394 Mt. FY2018 and FY2019 were previously absent, recovered from the 2020–21 edition's historical table; FY2025 is provisional by its edition's own declaration.

from 0.423 in FY2022 to 0.312 Mt in FY2023 before recovering. And it peaks in FY2019 at 0.477 Mt, 22 % above the reference year—a maximum that sat inside the two recovered years and was therefore invisible in every previous version of this figure. A single operator's annual fuel account is still a lumpy quantity, sensitive to service suspensions, locomotive availability and the timing of procurement and depot issues, and under the annual-equivalence assumption of Section 3 the series should be read as the operator's fuel account rather than as a smooth measure of traffic.

7.5. Provincial allocation

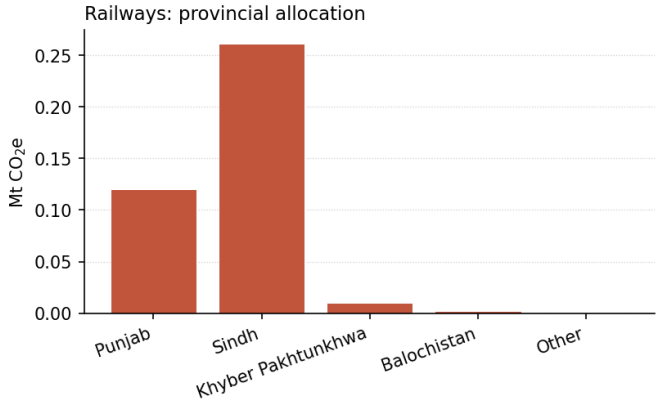


Figure 5: Reference-year administrative railway emissions. The current primary allocation and the separately licensed ODbL route alternative retain distinct geographic support; an administrative total is not proof of supported district or grid placement.

The ODbL route model and the loading-point proxy answer different allocation questions. Corridor service weights are conditional, and a freight-loading location does not identify the full route over which fuel is burned. Primary district/grid placement remains unallocated where a compatible physical rail surface is unsupported. A finer nominal grid does not remove the uncertainty in activity or service weights.

7.6. Place in the national inventory

At 0.09 % of the national total, railways is not a key category by level and contributes nothing measurable to national variance. It is included because completeness is a property of an inventory rather than of its largest categories, and because the attribution it settles—national diesel the road category once absorbed wholesale, now excluded from the road control at its source and verified by an independent two-source comparison—is roughly a third of a megatonne, the size of this entire sector.

8. Uncertainty

8.1. What a percentage can say

One metered account drives every stream, so the activity components of the row uncertainties are correlated; the grouped Approach 1 treats each fuel's activity term as perfectly shared across its gas rows and lands at ±14.7 % on the sector total, against 14.55 % if the rows were independent. The interval's shape is the sector's story in miniature: diesel CO_2 at ±5.4 % is among the best-constrained streams in the national inventory, while diesel N_2O carries ±150 % and is a tenth of the category—the sector is simultaneously the most accurately measured and, in its second gas, the least confidently factored. The full inputs register, with every quantity the statement rests on, ships as a generated output.

8.2. What a percentage cannot say

Discrete choices are named cases with conditional intervals and no probabilities attached, in this inventory's standard idiom:

Scenario	Mt CO_2e	What it answers
reference	0.39437	the estimate
road_n2o_factor	0.36282	the road table on rail—the CRT's own treatment
ie_summary	0.39437	the earlier summary booking restated: everything inside 1.A.3.c, the 0.00498 non-traction share declared IE; the reference itself now books the recode
stock_timing_prior_year	0.35323	consumed-equals-burnt bounded by the prior-year mean
tier2_service_split	0.39437	invariant by construction; the split moves nothing

Table 7: Structural scenarios for the greenhouse-gas total. The SO_2 sulfur ladder (0.008 kt to 2.25 kt) and the provincial wagons-proxy alternative are scope-noted separately in the generated scenario table; the once-live summary-vs-detail scenario is retired, its question settled by the 2024–25 edition's restatement.

The largest genuine uncertainty in the sector is not in its total at all. It is the N_2O factor's published range, which nothing short of measurement on Pakistani locomotives would narrow—and at 0.09 % of national emissions the case for that campaign is air quality and fleet performance, not greenhouse-gas accounting.

9. Discussion

9.1. When a submission can be refuted, and when it cannot

Independent checking of national greenhouse-gas submissions usually ends inconclusively. Two estimates differ; both are models; neither is a measurement; and the comparison records a disagreement without resolving it. This inventory's fugitives sector is a clean example, where a factor of 1.3 against EDGAR settles nothing because both estimates choose within the same published IPCC range and neither has observed a Pakistani gas system.

Railways is the opposite case, and the difference is worth stating precisely because it is not about rigour. It is about what the category rests on.

A fugitive emission factor is a statement about infrastructure nobody has measured. A railway fuel figure is a quantity a state enterprise metered, paid for and published, and the carbon in it is fixed by chemistry. When a submitted figure requires ten times that fuel, the disagreement is not between two models—it is between a figure and an audited quantity. And when a second, independent statistic in the same publication puts the implied fuel intensity at 46 L per train-kilometre, the refutation no longer depends on trusting the fuel account either.

The general form: a category can be refuted from outside when its activity datum is a metered physical quantity and a second published statistic bounds the same quantity independently. Railways has both. Most categories have neither, which is why this kind of result is rare rather than because verification is usually done badly.

9.2. What probably went wrong in the submission

This paper does not know how the BUR's 4.06 Mt was produced, and speculating about it is not the same as refuting it, so that refutation stands on the arithmetic alone. But the current CRT localises its own version of the error, because a CRT publishes intermediates the BUR did not: the activity cell is 12.6 times the metered fuel—a magnitude characteristic of a unit error, litres read as something larger—and the implied-factor cells show the road table where the rail table belongs. Two errors, separable in the filing's own arithmetic, each of which the operator's two published statistics would have exposed immediately.

The constructive point is procedural rather than diagnostic. Had the submission published its activity datum alongside its emissions, the disagreement would have been visible immediately and resolvable by anyone. It is the absence of the intermediate quantity, not the estimate, that makes the figure hard to check—and publishing implied fuel per train-kilometre alongside the emissions would cost nothing and make the category self-auditing.

9.3. What global inventories are missing, demonstrated

The ELK comparison is worth generalising. A careful global fleet model, given no Pakistani activity data, lands a factor of two low nationally and inverts the passenger–freight composition entirely—five grid cells of passenger rail for an operator that runs four passenger train-kilometres for every freight one. The two numbers that would have fixed it, annual fuel and train-kilometres, have been published in the operator's Year Book every year since the 1950s. The marginal value of nationally reported operator data to global inventories is usually asserted; here it is measurable, in both magnitude and composition.

9.4. The third instance of the parallel-table trap

Volume 2 Chapter 3 contains several near-duplicate default tables. Their CO_2 rows agree, because CO_2 depends only on the carbon in the fuel; their CH_4 and N_2O rows do not, because those depend on the engine and the duty cycle.

This inventory has now hit that structure three times: residential fuel against the power table, power against several, and rail against road here. In every case the failure would be silent—the dominant gas is unaffected, the fuel reconciliation still closes, and the error appears only in a minor gas at a magnitude that looks like ordinary uncertainty.

The defence that generalises is not vigilance. It is to publish the counterfactual. This sector computes what it would report on the wrong table, 0.363 Mt against 0.394, and writes it into its own output as a labelled diagnostic row. A reader can then see the size of the choice without re-deriving it, and a future edit that silently switched tables would move a number that is checked rather than one that is not.

9.5. Small categories and the attributions they settle

At 0.09 % of national emissions, railways would fail any threshold test for inclusion. Its value to the inventory is not its own magnitude but the attribution it settles: the national high-speed-diesel figure covers road and rail together, and before the road sector re-anchored on the petroleum industry's customer classes, the road category absorbed the whole of it. The two fuels are now separate rows at the source, and this sector's metered account is the independent check that the Railways row is right.

That is a general property of small transport categories. They are frequently the reason a large category's activity data can be stated correctly, and dropping them on grounds of size leaves the large category quietly wrong by the amount that was dropped. Rail was worth about a third of a megatonne to road transport, which is roughly the entire railway sector—the correction to the neighbour was the same size as the category making it, and it now lives in the source data rather than running as a deduction.

10. Limitations

The separate ODbL route alternative has mapped geometry and assumed corridor weights. The network, the yards and the route-kilometre audit are data. The service intensity assigned to each corridor is declared judgment, because Pakistan Railways publishes no service-frequency statistics and its public timetable interface refuses unauthenticated access; the basis of every weight is stated on its data row, the freight-wagons proxy is retained as the structural alternative, and the difference

between the alternatives is a structural sensitivity, not a confidence interval. A timetable extraction, or divisional train-km if any edition ever carries them, would replace the judgment with measurement. The national total is unaffected throughout.

Workshop and stationary fuel is inside the module's metered total, measured, and booked at its own category. Fuel burnt by railway workshops, power houses, pumps and track machines is part of the same consumption total this sector reads—the source says so itself: "these quantities comprise the entire fuel consumed on the Railways, i.e. on locomotives, power houses, water pumps, workshops etc."—and IPCC §3.4.1.4 flags the overlap explicitly. That fuel belongs outside 1A3c, but not in any single category. This paper previously assigned the whole quantity to 1A4c ii, which was a classification by purchaser rather than by source: under the IPCC definitions, track-maintenance machinery is off-road transport, workshop generators and power houses are stationary combustion, and pumps follow their end use. What remains unresolved is only that FINAL split: the measured non-traction quantity books at its own category (1A2/1A4c unresolved) rather than inside 1A3c, and no published record yet decomposes it across those end uses.

This paper previously also called that fuel inseparable. It is not. The same edition carries a detailed fuel table on p.163 splitting locomotive from other-than-locomotive consumption, for diesel and furnace oil, back to 1950—annually from 2015–16, in five-year averages before that—one page from the summary this sector was built on. Non-traction diesel is 1599 thousand litres, 1.21 % of the total (since the 2024–25 restatement below, the two tables' denominators agree). The overlap is small, which is what the sector had assumed; the difference is that it is now measured—and published as a ledger: `outputs/railways_fuel_claimants_2024.csv` carries each claimant with its volume, share and emissions at the sector's own factors, closing to the headline exactly. The recode IS the reference: the engine books 0.004 98 Mt at its own non-traction category and 1.A.3.c at locomotive traction, with the older summary booking preserved as the configuration *rail_ie_summary* in `outputs/national_configurations.csv`. Either booking conserves the national total exactly—the same fuel at the same factors under a different code, never a deletion. (An earlier edition priced this axis as a subtraction; the conserving recode replaced it, and the reference itself now carries the split.)

The Year Book disagreed with itself about the reference year, and its next edition settled it. Checking the summary against the detail across every year both tables cover produced exactly one failure: the 2023–24 edition printed the FY2023–24 locomotive cell as "136.988"—a decimal point where every other row in its column uses a comma—making the detail sum 138 587 against the summary's twice-stated 132 588, a 4.52 % gap. This paper reported the summary, on the strength of two statements against one typographically anomalous row, and published the detail reading as a labelled sensitivity of 0.412 Mt against 0.394.

The 2024–25 edition resolved it. Its historical run restates the cell as 130 988—printed with the same decimal-point quirk on the same row—and 130 988 + 1599 = 132 587, within one part in 132 588 of the summary. The misprint hypothesis was right, the reported number never moved, and the sensitivity is retired. The reconciliation now closes in every year:

FY	Summary	Loco	Other	Detail
2015–16	139 487	137 908	1 579	139 487
2016–17	144 631	142 607	2 024	144 631
2017–18	156 396	154 357	2 039	156 396
2018–19	159 927	157 979	1 948	159 927
2019–20	139 600	137 969	1 630	139 599
2020–21	140 257	138 841	1 416	140 257
2021–22	142 086	140 482	1 601	142 083
2022–23	104 909	103 272	1 637	104 909
2023–24	132 588	130 988	1 599	132 587
2024–25	138 673	137 096	1 577	138 673

Table 8: Thousand litres of diesel, across every year the two tables both cover. Eight reconcile to the digit and two to within three parts in 140 000; the FY2023–24 row carries the 2024–25 edition's restatement of the cell its predecessor misprinted as "136.988."

The check that found this ran late, and the honest reflection stands: this paper's argument is that a national submission can be refuted on a fuel account, and an inventory making that argument has to hold its own fuel account to the same standard. It now does, on every build—and the one anomaly the check ever surfaced was a misprint in a national statistical publication, identified here from the document's own internal evidence before the publisher restated it.

The furnace-oil basis is settled, and it was not what the summary said. Editions changed the published unit label without announcement, and this paper recorded the ambiguity as unresolved. The 2023–24 edition resolves it against itself: the p.163 column is headed "Furnace Oil (M. Tonnes)", the narrative on the same page states 79.907 M.T, and that is exactly the detail table's 15.083 locomotive plus 64.824 other. The Principal Statistics table calls the same quantity "Liters/thousand"—and its unit column is demonstrably unreliable, labelling Persons Employed "Tonne" and Cost of Employees "No." The basis is tonnes. The stream is under 0.001 Mt either way, as this paper already argued, so nothing material moves; the data now say the right thing.

N_2O is a tenth of the sector and carries ±150 %. The rail default's 95% interval spans 14.3 kg TJ^{-1} to 85.8 kg TJ^{-1}, which is the whole of the sector's uncertainty in practice. Nothing short of measurement on Pakistani locomotives would narrow it, and the sector is too small for that to be worth doing.

The air-pollutant layer rests on European fleet-average factors and a declared sulfur assumption. The EMEP/EEA Tier 1 locomotive factors describe an average European fleet, not Pakistan's; the chapter's own confidence intervals are carried on every factor row, and its printed intervals for the particulate rows do not bracket their central values—a defect of the source recorded on the rows rather than silently repaired. SO_2 depends entirely on fuel sulfur, for which no operator certificate is published: it is reported as explicit sulfur cases spanning 10 ppm to 10 000 ppm rather than as one number, with the pre-Euro-V 500 ppm specification as the declared reference.

Wear PM is exploratory and European-anchored. The wheel, rail and brake-wear layer (TNO (2024)'s fuel-based factors on the metered diesel: roughly 15 t of PM_{10} a year, a tenth of exhaust PM_{10}, alongside it and never summed) rests on Dutch and Austrian measurements. Pakistani conditions—tread brakes throughout, heavy axle loads, track condition—plausibly sit higher, the source's own $PM_{2.5}$ bullet contradicts its stated size fractions (recorded on the factor rows, the fraction-consistent reading adopted), and ±150 % is carried until a tread-brake-specific or Pakistani measurement exists.

One operator, one figure, no redundancy. The estimate has no second source. The only independent check the publication offers on itself is that the FY2023 figure is republished identically in two consecutive editions, and the traffic-based intensity check of Section 6. If the published fuel figure were wrong, nothing here would detect it.

11. Conclusion

Pakistan's railways emitted 0.394 Mt CO_2e in FY2023–24 — 0.389 Mt of it strict locomotive traction at 1A3c, 0.005 Mt the measured non-traction claimant at its own 1A2/1A4c-unresolved category — 0.09 % of a ten-sector national inventory, from 132 588 thousand litres of metered diesel on a mainline network that has been diesel-only since about 2010. Around that number this paper builds what the category never had: an air-pollutant layer (5.9 kt NO_x, 0.15 kt $PM_{2.5}$, an explicit sulfur ladder for SO_2), a twelve-year unbroken series with a quinquennial history to 1950 showing a furnace-oil-era peak 5.8 times today's total, a parametric-plus-structural uncertainty system, and a separately licensed ODbL route alternative. The primary release preserves its supported administrative allocation and explicit unallocated mass at finer grains.

The matched-year evaluation is the sharpest result. BUR-1's 4.06 Mt for FY2017–18 is 8.7 times the 0.466 Mt the identical method gives on that year's metered fuel; the current CRT books 12.6 times the metered activity with the road table's non-CO_2 factors readable in its own implied-factor cells; and the disagreements are pinned by tests that will fail—deliberately—the day a submission corrects them.

Four things generalise from a sector this small.

The strongest inventory architecture is a measured control plus reconciled decompositions. Rail shows the pattern at its purest: one metered number sets every magnitude; services, pollutants, geography and history are distributions over it, each reconciling exactly. Where agriculture needs six categories of structural assumptions, rail needs a fuel meter.

A submission can be refuted from outside when its activity datum is a metered physical quantity and a second published statistic bounds it independently—and publishing the counterfactual is a better defence against factor-table errors than care, because the parallel tables agree on CO_2 and diverge silently on everything else. The same check applies to the implied-factor cells of the national submission.

Operator data are what global inventories are missing, measurably. ELK, without Pakistani activity data, lands a factor of two low and models a passenger railway as a freight one. The fuel and traffic figures that would have fixed it have been published annually for seventy years.

Small categories earn their place by the attributions they settle. Railways was worth about a third of a megatonne to road transport, because the national diesel figure covers both and road would otherwise claim the whole of it; the split is built into the petroleum industry's own customer classes, and the metered account keeps the Railways row honest.

For Pakistan the recommendation is unchanged, narrow and cheap: publish railway emissions alongside the fuel and train-kilometre figures they were derived from. A category that shows its implied litres per train-kilometre audits itself, and neither the BUR's factor of ten nor the CRT's road factors would have survived a single edition.

Code and data availability

The calculation engine, the activity data and every output quoted here are version-controlled in the inventory repository (https://github.com/PakAirQuality/pakistan-emissions-sectors) and reproduce from a single command; Appendix A gives the sequence. Tests reconcile every table in this paper row by row against the engine's outputs, anchor each headline number individually, and recompute the derived arithmetic of Section 6; the remaining printed numbers are constants quoted directly from the primary sources, whose cached text extracts ship with the repository.

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Appendix A. Reproduction

```
pip install -e .
python3 -m pkinv.sectors.railways.run_railways
```

runs the whole stdlib pipeline and writes the 21 files its run manifest records as generated, closing with railways_run_manifest_2024.json, which lists them and hashes every primary source the run read. They are, by what each contains:

- the sector's own reported numbers: railways_totals_2024.csv (the sector total); railways_by_stream_2024.csv (the same by stream); railways_provincial_2024.csv (the provincial split); railways_monthly_2024.csv (the monthly profile); railways_timeseries_2024.csv (the twelve-year series in both GWP vintages); railways_fuel_2024.csv (fuel by stream in TJ and share).
- the air-pollutant layer: railways_air_pollutants_2024.csv (the pollutant totals); railways_air_sulfur_cases_2024.csv (the declared sulfur cases); railways_wear_pm_2024.csv (wear particulates by pollutant).
- the Tier 2 diagnostic: railways_tier2_split_2024.csv (the split itself); railways_tier2_air_diagnostic_2024.csv (its air-side diagnostic).
- the official comparison: railways_unfccc_crt_cell_audit_2024.csv (the CRT cell extraction); railways_official_comparison_2024.csv (this inventory against each official filing for that filing's own year).
- the uncertainty system: railways_uncertainty_2024.csv (the interval); railways_uncertainty_inputs_2024.csv (the input register behind it); railways_uncertainty_scenarios_2024.csv (the structural scenarios with their national delta).
- the reconciliations and quotients: railways_fuel_reconciliation_2024.csv (the summary fuel account against the locomotive and other-than-locomotive detail, with a verdict per year); railways_fuel_claimants_2024.csv (who else claims the same fuel); railways_efficiency_2024.csv (litres and MJ per train-kilometre, and grams of CO_2 per passenger- and tonne-kilometre); railways_qa_2024.csv (the quality-control report); railways_fuel_history.csv (the quinquennial history to 1950).

Three further tables come from phases outside the runner and so are absent from its manifest: railways_route_province_shares_2024.csv, the route-based provincial shares; railways_elk_crosscheck_2024.csv, which sets the ELK inventory's 2019 Pakistan cells against this one by species and subsector; and railways_gis_provincial_2024.csv, the provincial split computed from the rail network itself rather than from the route shares, carrying NO_x, $PM_{2.5}$ and SO_2 in kilotonnes beside the AR5 total. It comes from the spatial entry point behind the [gis] extra, so it is a held product with its release digest recorded rather than one the default build rewrites.

The spatial products need the gis extra:

```
pip install -e ".[gis]"
# from the Geofabrik archive; railways-gis then
# writes GeoTIFFs and the district tables
python3 scripts/build_rail_network.py
make railways-gis
```

Figures are regenerated with python3 -m pkinv.report.figures and synced by python3 scripts/sync_figures.py. The sector's invariants run with

```
python3 -m unittest discover tests \
  -p "test_railways*.py"
```

—about a hundred sector tests across six files—and the nineteen generic contract tests run over this sector as over every other. The manuscript itself is under test: every registered table is reconciled row by row against the CSV it was set from, and each headline number carries an anchored claim.

Appendix B. Sources

Fuel consumption, traffic and the detail tables are the Pakistan Railways Year Books (five editions held with cached text extracts; the scanned 2024–25 edition hand-transcribed from page images with page-cited provenance). Diesel density is Pakistan State Oil's high-speed-diesel material safety data sheet. Greenhouse-gas factors are IPCC 2006 Guidelines Volume 2 Chapter 3 Table 3.4.1 (rail) with net calorific values from Volume 2 Chapter 1 Table 1.2 (IPCC, 2006); air-pollutant factors are the EMEP/EEA Guidebook 2023 railway chapter (Norris and Ntziachristos, 2023), with the Tier 2 factors tracing to the UIC Rail Diesel Study (Halder and Löchter, 2005). The Railways customer-class row behind the two-source diesel comparison is the OCAC sectoral sales report, the same publication that anchors the road sector's controls. The official comparators are Pakistan's First Biennial Update Report and its Common Reporting Tables (extracted cell-by-cell from the archived workbook, SHA-256 pinned). The railway network derives from the OpenStreetMap Geofabrik extract (OpenStreetMap contributors, 2026) (ODbL 1.0; the extracted network and yard tables are derivative databases, attribution and audit in `data/gis/rail/`); the global comparator is the ELK land-transport inventory (Righi et al., 2026); traffic corroboration is the World Bank WDI railway series (UIC Railisa, CC BY 4.0); and the independent operational evidence is the Auditor-General of Pakistan's railway audit reports, para-cited in a versioned evidence register.